

29/07/26

ACTIVITY - 2

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1) $\text{push}(54); \text{push}(52); \text{pop}(); \text{push}(55); \text{push}(62);$
 $s = \text{pop}();$

consider the following sequence of operations
on an empty queue.

$\text{enqueue}(21); \text{enqueue}(24); \text{dequeue}(); \text{enqueue}(28);$
 $\text{enqueue}(32); q = \text{dequeue}();$ The value of $s + q$

Stack

$\text{push}(54) \rightarrow [54]$

$\text{push}(52) \rightarrow [54, 52]$

$\text{pop}() = 52$

$\text{push}(55) \rightarrow [54, 55]$

$\text{push}(62) \rightarrow [54, 55, 62]$

$s = \text{pop}() = 62$

$\therefore s = 62$

Queue:

$\text{enqueue}(21) \rightarrow [21]$

$\text{enqueue}(24) \rightarrow [21, 24]$

$\text{dequeue}() \rightarrow \text{removes } 21$

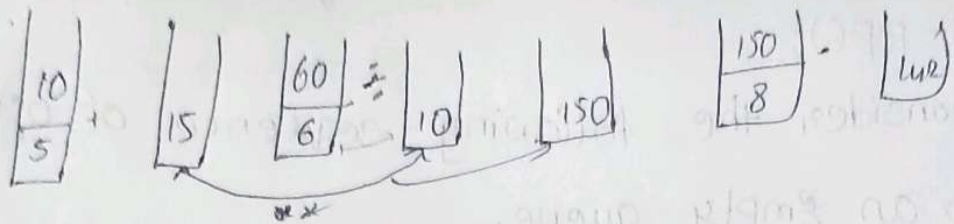
$\text{enqueue}(28) \rightarrow [24, 28]$

$\text{enqueue}(32) \rightarrow [24, 28, 32]$

$q = \text{dequeue}() = 24$

$s + q = 62 + 24 = 86.$

2) $10\ 5 + 60\ 6 / \times 8 -$



3) Minimum number of Enqueue operation

$Q_1 \rightarrow n$ elements

$Q_2 \rightarrow n$ elements

4) REVERSE operation

stack order $O(n)$

5) Postfix Expression

$8\ 2\ 3\ ^\wedge\ / \ 2\ 3\ ^\star + 5\ 1\ ^\star -$

After first $\star$ is evaluated the top two stack elements are 6, 5

6) $abc^\star + def^\wedge n^\star$

Infix to Postfix

[Signature]